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Multiphase model for nanoimprint lithography☆

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ABSTRACT

Reynolds' lubrication theory is extended to capture two-fluid dynamics based on relative permeability concepts used for multiphase flow in porous media. A reduced-order model of gas dissolution is constructed by combining two-phase lubrication theory with Henry's law and analytic approximations of surface area and diffusion length. The governing equations are solved as a thin-shell, finite-element model that can be used to investigate process sensitivities to a variety of technologically important parameters. Results are compared to those from related computationally intensive alternative formulations and to experimental flow visualizations to demonstrate its efficacy in modeling large area, many-drop processes and capturing the gas dissolution behavior observed in practice.

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1. Introduction

The patterning of surfaces with features smaller than 100 nm is of great importance to the printed electronics, semiconductor and related manufacturing industries, in which low-cost production of nanometer-scale featured surfaces is necessary to meet evolving market demands (Morse et al., 2011). Jet-and-flash imprint lithography (J-FIL) (Schmid et al., 2009), a form of ultraviolet (UV) nanoimprint lithography (NIL) (Stephen Y. Chou, 1995), has been rapidly advanced to meet these demands (Haisma et al., 1996). Pictured in Fig. 1, J-FIL involves the interaction between solutions of photoactive monomer and solvent in a liquid phase with a gas phase in a thin, closing gap between a substrate and a patterned template (Colburn et al., 2004).

Fig. 1a represents the first step of the J-FIL process in which 1000s of micro-size drops $\mathcal{O}$(1–10μm) are dispensed over relatively large areas $\mathcal{O}(10\,\mathrm{cm}^2)$ in non-uniform arrays, registered with the pattern to be imprinted, so as to distribute the liquid where it can efficiently fill the template with minimal waste. The combination of applied pressure and capillary forces exerted by the liquid bridges closes the gap between the surfaces as the drops merge and features fill, depicted in Fig. 1b. This process creates a two-phase flow of immiscible fluids in an exceedingly thin $\mathcal{O}(15\,\mathrm{nm})$ gap before the liquid fills its designated volume, depicted in Fig. 2a. Before the liquid is cured with UV light (Fig. 1c), ideally, all gas has been eliminated from the gap, the liquid has filled the template and the residual layer below the features is uniform over the print area Usuki et al. (2012). Once the pattern is set, the template is separated from it, leaving behind the nano-patterned surface (Fig. 1d). The model developed in this paper focuses on the drop merger process of the pattern fill step.

As this process has been used to produce patterned surfaces with feature half-pitches as small as 4 nm (Li et al., 2012), NIL has been recognized by the International Technology Roadmap for Semiconductors as a possible next generation lithography candidate (Schift, 2008). Despite significant potential for these processes to advance manufacturing efficiency and precision, mitigation of print-defects remains a significant challenge (Morse et al., 2011; Schift, 2008). Management of the liquid drop coalescence in the closing gap is critical to process optimization because trapped gas that does not dissolve into the liquid results in non-fill defects and long dissolution dwell times lead to reduced process throughput. This paper puts forth a mathematical formulation based on m flow concepts that aggregates the micro-size effects of many-drop merger over large areas of varying liquid density and gap thick-

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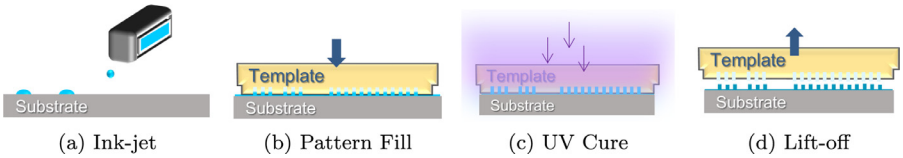


Fig. 1. Schematic diagram of jet-and-flash imprint lithography (J-FIL), integrable with wafer-scale/roll-to-roll processes. (a) Photoresist applied to substrate via precise ink-jet. (b) Photoresist drops merge and fill patterned template. (c) Photoresist is cured with ultraviolet light. (d) Template is separated from patterned template.

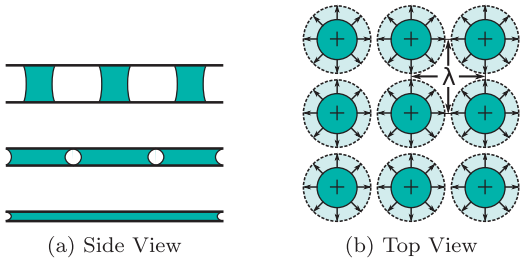


Fig. 2. (a) Side view of drops merging during pattern fill. (b) Top view of drop interface evolution during pattern fill.

ness. The mathematical formulation is solved with the finite element method and allows expedient simulation of the merger of thousands of drops over tens of cm^2.

Gas trapping and residual layer thickness (RLT) variations are key limiters of process rate and yield, as trapped gas impedes feature filling and non-uniform residual layers adversely effect subsequent etching processes. Physically consistent mathematical models are essential to the development of predictive process simulators. Simulator efficacy also depends on a robust and rapid numerical solution method so that a wide range of process parameters can be explored quickly.

Computational analysis of the drop merger regime (i.e. Fig. 1b) over areas containing hundreds to tens of thousands of drops is very challenging in part due to the multitude of interfaces and complex liquid-gas flow topology evolution illustrated in Fig. 2b. As the drops merge, filling of nano-size features in the template draws liquid away from the gap over much shorter time scales than those of dissolution of micro-size gas bubbles within the gap Chauhan et al. (2009). However, without liquid at the feature inlets the capillary forces don't exist to drive filling of those features. A reduced-order model that is predictive of gas trapping without resolving individual drop interfaces will enable large area simulations that examine residual layer thickness uniformity.

This study builds toward a complete predictive model to guide process designs that minimize gas trapping and reduce variation in the residual layer. Expedient model set up and short run times are also desirable so that design and scale-up of the process can be realized. At a minimum, the model requires first a suitable two-phase flow model, capable of scaling up to large flow areas while capturing the aggregate micron-scale effects without computational complexity. As such, trapping of micro-sized gas bubbles, subsequent gas dissolution and diffusion through the liquid phase are the focus of this study. The effects of capillary forces, structural mechanics and flow into template features are left for future work.

Efforts to model the J-FIL process began with work by Colburn et al. (2004). They assessed template control schemes with an analytic approach that accounted for both viscous resistance of the liquid phase and capillary forces as a function of gap thickness, gap closure velocity, liquid material properties and drop pattern. Reddy and Bonnecaze (2005) extended Colburn's analytic model with a numerical finite difference approach based on volume-of-fluid interface tracking to predict fill times for non-uniform drop patterns. Liang et al. (2007) implemented a gas bubble model with the finite difference method that predicts dissolution rate with Henry's law and Fickian diffusion to investigate how thermodynamic properties of the gas effect dissolution dwell times. Roberts et al. (2013) implemented a general two-phase curvilinear lubrication formulation suitable for non-planar flow domains typical of a J-FIL process with structurally flexible templates. Solving the resulting equations with a finite-element based approach, they simulated two-phase gap closure flow with level-set interface tracking. Moreover, they validated their results against analytic models for single drop squeeze flow and by accurately predicting evolution of the interface of a single drop as observed in experiment. Scale-up trials by the current authors to 100s and 1000s of drops using the approach taken by Roberts et al. revealed the method required too fine of a mesh to be practical for process design.

More recently several authors have addressed fluid management in related drop-dispensing, imprint lithography processes. A full three-dimensional, Navier–Stokes, two-phase mixture flow approach was demonstrated by Peng et al. (2016) to simulate micro-size bubble formation in micro-size features using the finite volume method. An innovative approach based on contact mechanics has been developed by Taylor (2016) that includes imbibition of liquid into the template and the motion of liquid across the substrate and demonstrated to scale to manufacturing areas. Jain et al. (2016) have advanced a finite-difference, volume-of-fluid (VOF) approach to address the J-FIL process that allows for large area, ($\mathcal{O}(1000)$) drop simulations, but ignores the gas phase. Nonetheless, he provides an analytical model that predicts an NIL parameter envelope wherein gas physics are negligible (Jain et al., 2017).

While useful process understanding has been gained from these works, no approach accounts for the effects of gas trapping and pressure-driven gas dissolution with a continuum model over practical processing areas of $\mathcal{O}(10\,cm^2)$. The approach put forth in this paper captures gas trapping and gas dissolution in a multiphase flow formulation that enables large scale (>1000 drops) simulations suitable for iterative process design studies. The continuum approach is adapted from analyses of the Saffman-Taylor instability in oil-water flows through fractured rocks (Fourar et al., 1992) in which a two-fluid extension of Darcy flow with *relative permeability* (Wyckoff and Botset, 1936; Lockhart and Martinelli, 1949) is employed. The formulation is developed in Section 2 utilizing multiphase concepts of saturation and relative permeability augmented by a reduced-order gas dissolution model. Section 3 presents model results, demonstrating simple verification and one envisioned application. Comparison between simulation and experimental flow visualization demonstrate the model captures two-phase phenomena pertinent to prediction of gas dissolution dwell times. Section 4 discusses the results and future work.

2. Model development

In this section, a mathematical model is developed to capture two-phase flow evolution from the moment at which the drops bridge the gap between substrate and template to beyond the point at which all of the gas has either escaped or been dissolved into the liquid, as depicted in Fig. 2a. Fig. 4 shows snapshots

from an experimental flow visualization of the process as captured with a high-speed camera. Model development follows observations that as the gap thickness is decreased, flow is separated into three stages by the events pictured in Fig. 4. Formation of liquid bridges between upper and lower surfaces marks the beginning of the first stage in which gap volume decrease occurs by gas phase displacement. Contact of the drop edges with one another (drop merger) marks the beginning of the gas compression stage. During this stage, viscous forces in the thinning gap lead to large pressure gradients in the liquid as it displaces the compressing gas. Pressure builds up in both gas and liquid phases to offset the squeeze forces, reducing the gap closure rate and initiating the gas dissolution stage. During this stage, long drop merger times due to increased resistance to liquid flow in very thin gaps may be exacerbated by the requirement that trapped gas must dissolve into the liquid to avoid defects. Previous findings that both smaller Henry's Law constants (Liang et al., 2007) and smaller drops (Peng et al., 2016) lead to shorter processing times motivates the inclusion of gas effects in this coarse-grain model. Work demonstrating that dwell times can be practically eliminated with condensible gases (Hiroshima et al., 2013) juxtaposed against findings of increased shrinkage with condensible gases (Wang and Hiroshima, 2010) illustrates that gas phase management plays a critical role in the process and drives model development towards the study of gas dissolution.

An idealized model is developed in the remainder of this section assuming drops of identical volume, V_d, are arrayed in a uniform grid across the substrate in a square lattice with a pitch of λ, as shown in Fig. 2b. Actual applications often require array density variation over the device print area, as discussed in Section 3.4.

2.1. Lubrication flow

The small height of standing drops, $\mathcal{O}(1\mu m)$, and large areas necessary for manufacturing processes $\mathcal{O}(10cm^2)$ result in a flow domain of extreme aspect ratio, making a full three-dimensional multiphase flow model impractical. Reynolds' lubrication approximation (Reynolds, 1886) simplifies the Navier-Stokes equations, dropping negligible terms when the gap thickness, h, is small compared to the flow length, L, and the product of the ratio of inertial to viscous forces (Reynolds number Re) with the ratio of gap thickness to flow length is small ($Re \cdot h/L \ll 1$) (Deen, 1998). Stokes (Stokes, 1898) integrated the resulting three dimensional equations through the gap thickness assuming the surfaces have no relative, in-plane motion. The resulting expression for the gap averaged velocity, $\bar{\boldsymbol{v}}$, as a function of the pressure gradient, ∇P, is used still to describe two-dimensional, streamline flows in Hele-Shaw cells (Hele-Shaw, 1898).

$$\bar{\boldsymbol{v}} = -\frac{h^2}{12\mu}\nabla P. \tag{1}$$

Here, P, is the fluid (or lubrication) pressure, h is the gap thickness and μ is the dynamic viscosity.

2.2. Mass balance

An approach described by Brennen (2005) as *disperse*, two-fluid flow is taken in this development because it inherently allows multiple fluids to occupy the same space and avoids the need to track explicit interfaces. Instead of tracking motion of two fluids *separately*, the volume fractions are tracked by the "saturation" in the gap, where a diffuse representation of the liquid volume is used as depicted in Fig. 3.

Continuity of two immiscible, yet reciprocally dispersed fluids is satisfied by employing two mass balance equations, one for each phase, adapted from phase field models of Hele-Shaw flow

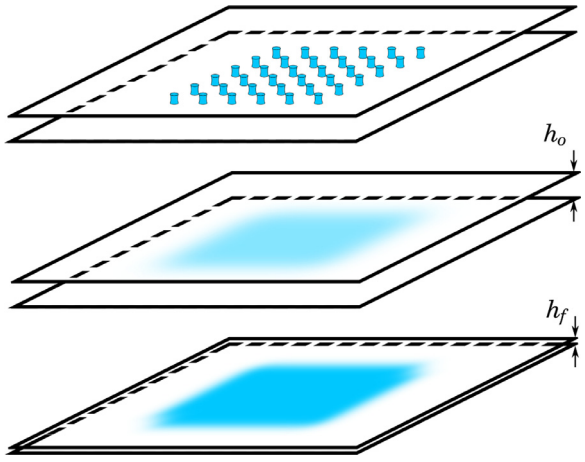


Fig. 3. Ink-jetted drops represented as a diffusely saturated patch.

(Cueto-Felgueroso and James, 2014). Each mass balance is integrated through the gap thickness and makes use of gap average velocities, yielding planar mass conservation equations.

$$\frac{\partial(h\rho_l S)}{\partial t} + \nabla\cdot(h\rho_l \bar{\boldsymbol{v}}_{\boldsymbol{l}}) = 0 \tag{2}$$

$$\frac{\partial(h\rho_g S_g)}{\partial t} + \nabla\cdot(h\rho_g \bar{\boldsymbol{v}}_{\boldsymbol{g}}) = 0. \tag{3}$$

Here, t is time, ρ_l is the bulk density of the liquid phase, ρ_g is the bulk density of the gas phase and $\bar{\boldsymbol{v}}_{\boldsymbol{l}}$ and $\bar{\boldsymbol{v}}_{\boldsymbol{g}}$ are the in-plane, gap-average velocities for the liquid and gas, respectively. The liquid volume fraction, S, or saturation, together with the gas volume fraction, S_g, make up the whole imprint volume, viz. $S + S_g = 1$. To capture two-phase behavior, the concept of relative permeability is used to transform Eq. 1 into two separate velocity expressions, one for each phase.

2.3. Relative permeability

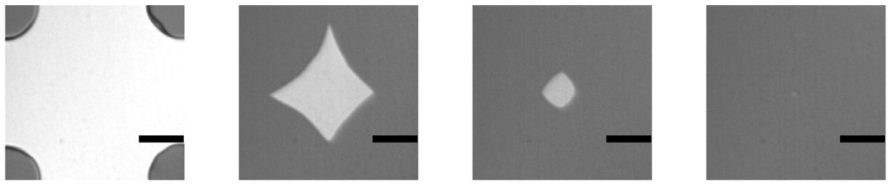
The concept of permeability is extended from models of flow in porous media, in which volume flux density, $\boldsymbol{q}$ (dimensions: length/time), is proportional to the pressure gradient:

$$\boldsymbol{q} = -\frac{k}{\mu}\nabla P \tag{4}$$

The permeability, k, lumps the effects of pore size, tortuosity, void fraction and level of saturation into a single, but variable parameter. Two-phase flow in an open channel between two flat plates is analogous to flow with a single but variable pore size i.e. the gap thickness. The similarity between Eqs. 4 and 1 suggests that the permeability of a thin gap is $h^2/12$.

Another concept borrowed from modeling approaches to flow in porous media is an additional factor of "relative permeability", used to describe the flow of one phase as it is impeded by the other phase(s) (Bear, 1972). In other words, the flow of one phase, as driven by the pressure gradient, is restricted by the presence of the other phase. Therefore the relative permeability depends directly on the saturation and the nature of interaction between phases. Mathematically, the relative permeability k_α of phase α relates superficial velocity[1] $\bar{\boldsymbol{v}}_{\boldsymbol{s\alpha}}$ of phase α to effective volume flow

[1] The superficial velocity of a phase is the velocity computed as though that phase were the only phase present as it would be driven by the pressure gradient.



(a) t = 0 ms : Drops bridge gap (b) t = 220 ms : Drops trap gas (c) t = 240 ms : Gas compresses and dissolves (d) t = 1.26 s : Gas phase eliminated

Fig. 4. Stills captured from high-speed video of drops squeezed between flat plates with 40 μm scale bar. (a) The drops bridge the gap between substrate and template. (b) The drop edges contact one another, trapping a gas pocket. (c) The gas pocket compresses under pressure forces imparted by both applied squeezing and surface tension driven suction between the substrate and template. (d) The gas phase has been completely eliminated via dissolution into the liquid volume.

velocities, $\bar{v}_\alpha$ (Lockhart and Martinelli, 1949). In two-phase Hele-Shaw flow, the relative permeability augments the superficial velocity,

$$\bar{v}_\alpha = k_\alpha \bar{v}_{s\alpha} = \frac{h^2}{12\mu_\alpha} k_\alpha \nabla P. \tag{5}$$

Existing relative permeability models in the literature for Hele-Shaw flows address pressure-driven flow through fixed, wetted gaps (Bacri et al., 1992), or flows in which mixtures have randomly distributed interfaces (Fourar et al., 1998). In contrast, the J-FIL process is characterized by phase mixtures with structured distribution and evolution of interfaces within a closing gap between dry surfaces(i.e. squeeze flow).

Development of relative permeabilities that capture the J-FIL flow configuration is motivated in part by discussion of the gas displacement and gas compression/dissolution stages (demarcated by the frames displayed in Figs. 4a, 4b and 4d[2]). From the point the liquid has bridged the gap, the liquid phase is completely discontinuous throughout the domain, so liquid flow is limited to expansion of the drops. In fact, the experimental observations in Fig. 4 clearly suggest that net liquid flow across the lattice boundary is negligible up to the point at which the gas phase is completely eliminated.

The gas phase appears to flow freely, past the liquid bridges, until it is trapped. At that point, the gas becomes the discontinuous phase in which no net macroscopic flow can take place. While Romm's experiments (Romm, 1966) find linear relationships between saturation and relative permeability for unstructured flow, drop-merger observations show that net flow of gas stops before the gap is fully saturated. Reconciliation of these observations is achieved in the model by using a tunable relative permeability model (constructed from piece-wise linear segments in saturation, S).

$$k_{rg} = \begin{cases} 1 & : 0 \le S \le S_{cg} - \beta_g \\ m_g S + c_g & : S_{cg} - \beta_g < S < S_{cg} + \beta_g \\ 0 & : S_{cg} + \beta_g \le S \le 1 \end{cases} \tag{6}$$

$$m_g = -1\frac{1}{2\beta_g} \tag{7}$$

$$c_g = m_g(S_{cg} + \beta_g) \tag{8}$$

$$k_{rl} = \begin{cases} 0 & : 0 \le S \le S_{cl} - \beta_l \\ m_l S + c_l & : S_{cl} - \beta_l < S < S_{cl} + \beta_l \\ 1 & : S_{cl} + \beta_l \le S \le 1 \end{cases} \tag{9}$$

$$m_l = 1\frac{1}{2\beta_l} \tag{10}$$

[2] Dr. Shrawan Singhal of the University of Texas at Austin, personal communication

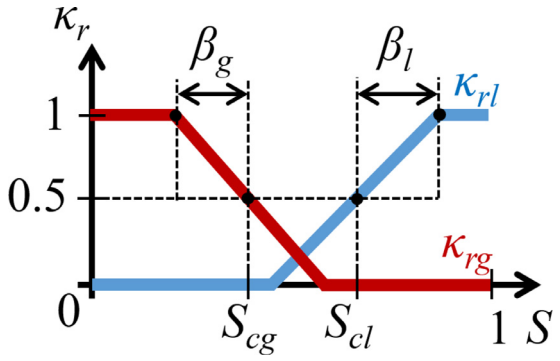


Fig. 5. Parameterization of two-phase permeabilities using transition centers, $S_{c\alpha}$, and half-transition spans, β_α, for liquid and gas phases.

$$c_l = -m_l(S_{cl} + \beta_l) \tag{11}$$

Here, the relative permeability for each phase, $k_{r\alpha}$, is tuned by selection of a transition center saturation, $S_{c\alpha}$, at which transition from free flow to no flow occurs and a half-transition length, β_α. These curves are depicted graphically in Fig. 5. The choice of these tuning parameters is influenced by the type of simulation being performed (diffuse patch of drops or discrete drops) and is discussed in Section 3.

2.4. Gas dissolution

The derivation of Eq. (1) assumes the fluid is incompressible, yet gas phase compression and resulting pressure rise is a driving force for its dissolution. A compressible gas model is introduced into the system via the approach used in studies of compressible lubrication flow (Harrison, 1913; Almqvist and Dasht, 2005). Simply, the gas density is computed according to the ideal gas law,

$$\rho_g = \frac{PM}{RT}. \tag{12}$$

Here, M is the molecular weight of the gas, R is the ideal gas law constant and T is the temperature, which for the purposes of this study is assumed to be constant (subsequent numerical demonstrations use $T = 298$K). Tracking pressure of the compressible gas phase allows the prediction of volume reduction by compression and dissolution into the liquid. The liquid is still taken to be incompressible, allowing for the liquid density to be divided out of Eq. (2).

The final stage of the process, seen beginning in Fig. 4c, is dominated by gas dissolution, a key rate limiting step. The liquid phase is taken here as an infinite sink that is initially saturated with gas at standard ambient pressure. Henry's Law states that the concentration of a chemical species, dissolved in liquid, is directly proportional to the partial pressure of the gaseous phase of that species,

i.e.

$$PH = b_g \tag{13}$$

Henry's law constant, H, relates the partial pressure of the gas species to the interfacial concentration of dissolved gas, b_g. The model assumes a single gas species, so the partial pressure is taken as the lubrication pressure, P. The magnitude of the diffusive flux density, j, of gas species into the liquid drop is modeled with a lumped mass transfer expression,

$$j \cong \frac{D_{gl}}{L}(b_g - b_\infty) = k_D(b_g - b_\infty), \tag{14}$$

where D_{gl} is the molecular diffusivity of the gas species in the liquid solvent, L is the diffusion length, b_∞ is the concentration of the dissolved gas species in the delivered photo-resist and k_D is the lumped mass transfer parameter. The diffusion length is approximated with an analytic expression based on the square grid drop pattern.

The volume-average diffusive flux of gas into the liquid phase, J is derived using the volume-average interface density, A_i/V_t, and the assumption that the magnitude of the flux density, j, is uniform across the gas-liquid interface,

$$J = j\frac{A_i}{V_t}h = h\frac{A_i}{V_t}\frac{D_{gl}}{L}H(P - P_{\text{atm}}). \tag{15}$$

Expressions for diffusion length and volume-average interface density are derived from the ordered evolution of the drop topology during gap closure. Before the drops contact each other, the diffusion length is taken as the radius of the drop, r, as defined by the drop volume, V_d, and the gap thickness, h,

$$L = r = \sqrt{\frac{V_d}{\pi h}}. \tag{16}$$

The volume-average interface density can be derived from the interface evolution of the square drop lattice. For every lattice element (Fig. 2b) with lattice length, λ, the interfacial area, A_i, is that of a single, cylindrical drop and the total volume is of the lattice element,

$$\frac{A_i}{V_t} = \frac{2\pi rh}{\lambda^2 h} = \frac{2\pi r}{\lambda^2}. \tag{17}$$

Because the saturation is related to gap thickness through the pitch and drop volume, viz.,

$$S = \frac{V_d}{V_t} = \frac{V_d}{\lambda^2 h}, \tag{18}$$

the volume-average interface density can be derived as a function of saturation by combining Eqs. 16, 17 and 18,

$$\frac{A_i}{V_t} = \frac{2\sqrt{\pi}}{\lambda}\sqrt{S}. \tag{19}$$

These expressions hold in areas of constant lattice density throughout the gas-flow stage, however once the drops begin to merge, their shapes no longer match a growing circle and the evolution of the two phase geometry becomes more complex.

A simplified geometry of the drop structure is used to calculate diffusion length and volume-average interfacial area after the gas is trapped. The gas pocket between the drops, pictured in Figure 6 b, resembles a shrinking square depicted in Fig. 6c throughout the gas-dissolution stage. The diffusion length is approximated to be the shortest distance, L, from the gas pocket to the drop center. Several relationships necessary to compute the volume-average interface density and diffusion length are derived from this simplified geometry. A structure parameter, y, the liquid area, A_l, and the quarter interface length, L_{qi}, of the quarter-lattice-element, depicted in Fig. 6d, relate the liquid drop volume and the lattice length to the inter interfacial area, diffusion length, saturation and gap thickness.

$$\frac{V_d}{4h} = A_l = \left(\frac{\lambda}{2}\right)^2 - \frac{y^2}{2} \tag{20}$$

$$L = \frac{1}{\sqrt{2}}(\lambda - 2y) \tag{21}$$

$$\frac{A_i}{4h} = L_{qi} = \sqrt{2}y \tag{22}$$

Re-organizing Eqs. 18, 20 and 21 yields the expression for gas-diffusion stage diffusion length in terms of gap thickness,

$$L = \frac{\lambda}{\sqrt{2}}\left(1 - \sqrt{2}\sqrt{1 - \frac{V_d}{\lambda^2 h}}\right). \tag{23}$$

An expression for the volume-average interfacial area in terms of saturation is found by combining Eqs. 18, 20 and 22.

$$\frac{A_i}{V_t} = \frac{4}{\lambda}(\sqrt{1 - S}) \tag{24}$$

The transition point between flow and dissolution stages occurs at the moment that drop edges make contact and trap gas pockets. This occurs at the moment when the drop radius is equal to half of the drop pitch. Therefore, the transition saturation, S_t, for the square drop lattice is equal to $\pi/4$.

With the assumption that the liquid is an incompressible fluid and combining the dissolution rate, J, into Eq. 3 as a sink term, the model equations become,

$$\frac{\partial Sh}{\partial t} + \nabla\cdot(\vec{v}_l h) = 0 \tag{25}$$

$$\frac{\partial \rho_g(1 - S)h}{\partial t} + \nabla\cdot(\vec{v}_g \rho_g h) + J = 0. \tag{26}$$

2.5. Numerical implementation details

The model Eqs. (25) and (26) and associated boundary conditions are discretized in space with the finite element method. The resulting nonlinear algebraic equations are solved with Newton's method and integrated in time with the backward Euler method. This numerical system is implemented in the multiphysics finite element software Goma 6.0 (Schunk et al., 2013)[3]. A Dirichlet boundary condition on pressure is sufficient when liquid does not cross the external boundary, however integration by parts of the advection terms in the weighted residual results in the default condition on saturation $\hat{n}\cdot(\boldsymbol{v}_l h) = 0$ at the boundary, where $\hat{n}$ is the vector normal to the boundary and in the plane of flow. The outflow boundary condition described by Papanastasiou et al. (1992) is used if liquid should exit the domain through the boundary. As the model was developed, mass lumping, a small amount of artificial diffusion and a conservative clipping method were found necessary to achieve stability of the transient simulations.

Reformulating the time derivative matrices that result from the FEM discretization with the mass-lumping (Zienkiewicz and Taylor, 2000) procedure mitigates unbounded growth of the pressure field near regions of sharp saturation transition. Still, spurious oscillations in the vicinity of these regions cause the saturation to overshoot the physically maximum value of unity Bradford and Katapodes (2000). A modulated artificial diffusion term, $D_a k_d \nabla^2 S$, is added to Eq. (25) to mitigate these spurious oscillations. This diffusion is tuned via the parameters β_d and S_{cd} as in Fig. 7. No

[3] https://goma.github.io.

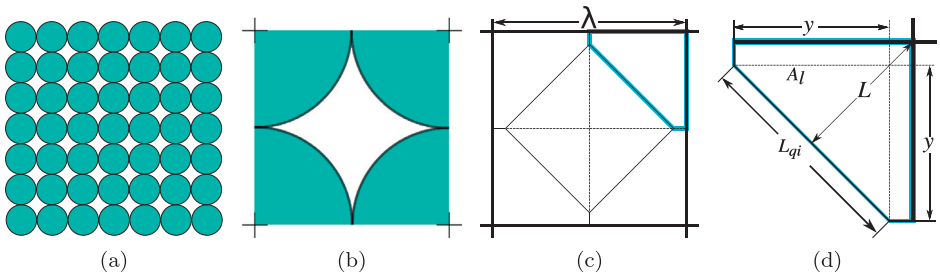


Fig. 6. Gas-dissolution stage model geometry (a) Square lattice of circular drops at point of drop-drop contact. (b) Square lattice element (c) Simplified 'diamond' geometry (d) Micro-structure geometry.

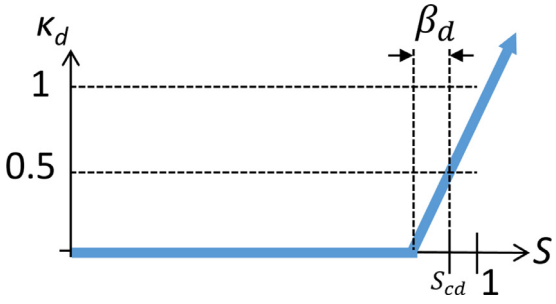


Fig. 7. Artificial diffusion modulation scheme.

mathematical constraints exist in the model to prevent the saturation from rising above unity and because the advection of each phase is tightly coupled to the level of saturation, these spurious fields result in increasing non-physical behavior as solutions evolve in time. Specifically, when the saturation is greater than unity, the gas phase mass balance, Eq. (26), drives the saturation higher. This non-physical behavior is remedied with a computational switch that substitutes the gas equation with an equation that acts to drive saturation back towards unity.

In regions devoid of gas, Eq. (26) as written serves no purpose, and so is replaced with one that promotes monotonicity of the simulation. When $S \geq 1$, the advection and dissolution terms go to zero, the time derivative is expanded by the product rule and $(1-S)$ is replaced with $k_c(1-S)^2$ to obtain,

$$0 = h\frac{\partial S}{\partial t} - k_c(1-S)^2\frac{\partial h}{\partial t} \tag{27}$$

Here, k_c is a parameter (chosen to be positive) that controls the strength of this spurious oscillation correction. Note that Eq. (27) applies correction by requiring $\frac{\partial S}{\partial t} < 0$, because $(1-S)^2$ is always positive and the closing gap implies $\frac{\partial h}{\partial t} < 0$. Since the liquid conservation equation (25) remains in effect, the conservation of the liquid volume is always maintained. Together, the modulated artificial diffusion and Eq. (27) constitute a form of "clipping" of the non-monotonic behavior (Leonard and Niknafs, 1991) of the saturation front. This clipping method has improved fluid tracking in thin-gap flow simulations, yet it is not sensitive to the change in gap thickness or variation in gap closure rate, making it difficult to tune for simulations with wide variations in gap thickness and closure rate.

3. Results

In this section the model is demonstrated at two levels, first on diffuse drop patches and then on discrete drop patterns. An energy-based approach to determining the specific relative permeability parameters of Eqs. (6)–(11) is discussed in Section 3.1 followed by discussion of sensitivity to key thermophysical parameters. A fine-grain, single-drop simulation exploring the limits of accuracy of interface evolution and pressure field predictions demonstrates that gas-liquid interfaces can still be captured effectively in Section 3.2. A simulation that captures the drop merger process is compared with experimental flow visualizations in Section 3.3. Finally, in Section 3.4, a coarse-grain simulation of varying density drop arrays is used to illustrate the class of problems this model was designed to support, and that the model is suitable for further advancement in subsequent work.

3.1. Diffuse drops

The manufacturing-scale fluid mechanics resulting from the merger of thousands of drops can be examined with coarse grain simulations by starting with a uniform saturation, representative of the NIL process. A patch of ink-jetted drops is represented by a saturation field with a value of the total volume of drops divided by the total gap volume, illustrated in Fig. 3.

For a diffuse representation of the regular square lattice depicted in Fig. 2b, liquid relative permeability can be constructed to maintain the expected viscous dissipation. To maintain the expected dissipation within a lattice element, the rate of work done on the fluid needs to equal the rate of dissipation due to irreversible conversion from kinetic to internal energy, as in

$$\int_A P\mathrm{d}A\frac{\partial h}{\partial t} = \int_{V_d} \boldsymbol{\tau} : \nabla \boldsymbol{v}\mathrm{d}V_d. \tag{28}$$

Here, A is lattice element area, V_d is the volume of the liquid within the lattice element, $\boldsymbol{\tau}$ is the viscous stress tensor and $\nabla \boldsymbol{v}$ is the velocity gradient.

In this approach, the notion of effective liquid pressure, P_l, is developed to relate forces developed due to drop liquid motion to coarse-grain lubrication pressure, P, and relative permeability. The motion of liquid is driven by the effective liquid pressure gradient which is equal to the gradient of the lubrication pressure modulated by the relative permeability,

$$\tilde{\boldsymbol{v}}_{\boldsymbol{l}} = -\frac{h^2}{12\mu_l}k_{rl}\nabla P = -\frac{h^2}{12\mu_l}\nabla P_l. \tag{29}$$

From Eq. (29), the assumption that the relative permeability is constant across a lattice element results in the relationship between lubrication pressure and effective liquid pressure,

$$k_{rl}P = P_l. \tag{30}$$

The effective liquid pressure is computed by dividing the total force per drop, F, by the lattice area.

$$P_l = \frac{F}{\lambda^2} \tag{31}$$

The force per drop is calculated by integrating the analytic expression for pressure field over the drop area, obtained by Reddy and Bonnecaze (2005).

$$F = -\frac{3}{2}\pi\mu_l\frac{r^4}{h^3}\frac{\partial h}{\partial t} \tag{32}$$

Now, the lubrication pressure changes negligibly across the lattice area so the rate of work done on the liquid can be computed as

$$\int_A P \mathrm{d}A \frac{\partial h}{\partial t} \approx P\lambda^2 \frac{\partial h}{\partial t}. \tag{33}$$

Breaking this expression into the sum of components due to dissipation in liquid and in gas gives

$$P\lambda^2 \frac{\partial h}{\partial t} = [P\pi r^2 + P(\lambda^2 - \pi r^2)] \frac{\partial h}{\partial t}. \tag{34}$$

Because the dissipation of energy comes predominately from the liquid phase, the second term on the right is neglected, giving

$$\int_A P \mathrm{d}A \frac{\partial h}{\partial t} \approx P\pi r^2 \frac{\partial h}{\partial t}. \tag{35}$$

Combining Eqs. 30, 31, 32 and 35 gives

$$\int_A P \mathrm{d}A \frac{\partial h}{\partial t} \approx -\frac{3}{2}\pi \mu_l \frac{r^4}{h^3} \left(\frac{\partial h}{\partial t} \right)^2 \frac{\pi r^2}{k_{rl}\lambda^2}. \tag{36}$$

Irreversible conversion from kinetic to internal energy is obtained using analytic expressions for velocity fields as presented by Chauhan et al. (2009) resulting in

$$\int_{V_d} \boldsymbol{\tau} : \nabla \boldsymbol{v} \mathrm{d}V_d = -2\pi \mu_l \left(\frac{3}{h^3} \frac{\partial h}{\partial t} \right)^2 \left(\frac{r^2 h^5}{5} + \frac{r^4 h^3}{12} \right). \tag{37}$$

Equating the two expressions for viscous dissipation, Eqs. (36) and (37) and simplifying, results in

$$\frac{1}{k_{rl}} = 12 \frac{\lambda^2}{\pi r^2} \left(\frac{h^2}{5r^2} + \frac{1}{12} \right). \tag{38}$$

In typical NIL processes, $h^2/r^2 \ll 1$, this allows removal of the first term on the right-hand side of Eq. 38. Furthermore, the saturation is the ratio of the liquid drop volume to the lattice element volume,

$$k_{rl} \approx \frac{\pi r^2 h}{\lambda^2 h} = S. \tag{39}$$

This relative permeability is achieved by selecting $S_{cl} = \beta_l = 1/2$.

As for relative permeability of gas, experimental observations show that gas no longer flows when the drops contact one another, i.e. when $S = \pi/4$. Because the gas is much less viscous than the liquid, its viscous dissipation can be neglected. Without regard to any viscous coupling of the phases or the resultant multiphase dissipation effects, a relative permeability that satisfies the flow/no-flow conditions for the gas phase results when $S_{cg} = \beta_g = \pi/8$. The sum of these relative permeabilities is less than unity except where $S = 0$ and $S = 1$, a result that is consistent with the experimental measurements of two-phase realtive permeabilities for a fracture taken by Fourar and Bories (1995).

Sensitivity analysis of a diffuse carbon dioxide-water system in a gap closing at a constant rate shows that increasing the number of drops to deliver the same liquid volume requires smaller applied force. In this model, smaller forces are allowed by faster gas dissolution resulting from decreased diffusion lengths and increased interfacial surface area (c.f. Eq. 15). The trend of this prediction is in agreement with lubrication studies of Colburn et al. (2004) and Reddy and Bonnecaze (2005).

A set of simulations is performed to demonstrate that the choice of gas species can have an effect comparable to an accessible change in the liquid viscosity. In each simulation, a patch of drops, saturated with the respective gas at atmospheric pressure, merge as the gap closes from 1μm to 100nm in 1ms. The base case simulation is that of the merger of water drops with a viscosity of 1e-3Pa·s in an atmosphere of nitrogen gas. Because the simulations are gap-thickness controlled and the dwell times are not an output of the model, a pseudo-work objective function is constructed to asses the energy requirement, E, corresponding to each test case. Specifically the pseudo-work is computed as,

$$E = \int_0^{t_f} \int_A P \mathrm{d}A \frac{\mathrm{d}h}{\mathrm{d}t} \mathrm{d}t. \tag{40}$$

Here t_f is the final time and A is the area of the simulation domain. The energy requirement is reduced by 9% when the liquid viscosity is reduced by 20% from the base simulation. In comparison, the energy requirement falls 8% when the gas species is switched from nitrogen in the base case to carbon dioxide. These comparable reductions in energy for independent changes in liquid viscosity and ambient gas species indicate that choice of gas species impacts processing rate just as much as liquid viscosity.

Table 1
List of physical parameters defining the simulations of a single drop in a closing gap.

Simulation Parameter	Value	Units
Initial Gap Thickness	1.32	μm
Initial Drop Volume	2.59	pL
Gap Closure Rate	1.36	$\frac{\mathrm{mm}}{\mathrm{s}}$
Simulation Time	800.	μs
Gas Viscosity	18.6e-3 (18.6e-6)	cP (Pa·s)
Liquid Viscosity	10.0 (10e-3)	cP (Pa·s)
Boundary Pressure	101	kPa
Domain Edge Length	250.	μm
Elements per Domain Edge	120	

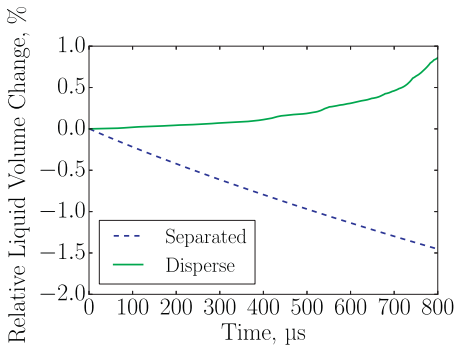
3.2. Single discrete drop

The disperse, two-fluid model was formulated to capture the effects of disperse, immiscible fluid flows without capturing the interfaces, however it can also be used to model flow of fluids that are separated by regions of stark saturation contrast, much like distinct drops. Numerical investigations of evolution of a single drop, flowing outward in a closing gap, as depicted in Fig. 2b, are reproduced with a level-set based, separated, two-fluid method (Roberts et al., 2013) and compared (Fig. 8) with simulations performed with the disperse, two-fluid model developed in this paper. A direct comparison of the two-phase lubrication models is achieved by excluding surface tension forces, taking gas density to be constant and foregoing the gas dissolution term, since the models do not possess complementary facilities for simulating them. The other relevant parameters are specified identically, as listed in Table 1.

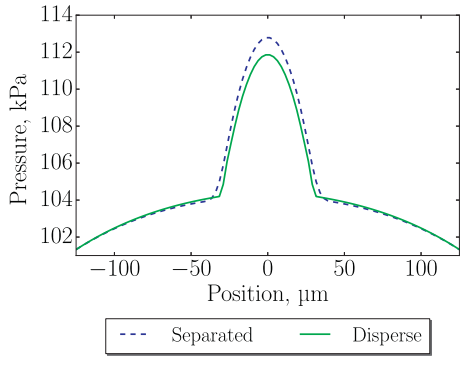
To model discrete drops as islands of fully saturated volume, relative permeability parameters are selected that facilitate neighboring flows with starkly different material properties. Transition centers and transition lengths of $S_{cl} = S_{cg} = \beta_l = \beta_g = 1/2$, represent a boundary symmetry in that each material interacts with the interface in the same manner. Figure8 c shows that increasing the interface resolution, either with denser mesh ($\epsilon = 240$) or smaller transition lengths ($\beta_l = \beta_g = 1/10$) leads to more accurate pressure predictions at later simulation times.

Fig. 8a shows the liquid volume as normalized by the initial liquid volume, computed via integration over the domain at each time step. The liquid volume error accumulation over time in the separated and disperse methods are 1.4% and 0.8% respectively. The pressure fields are compared at two different time steps in Figs. 8b and 8 c. The predicted pressure fields at 400μs are nearly identical, however, at the end of the simulation, the maximum pressure differs by nearly 20%.

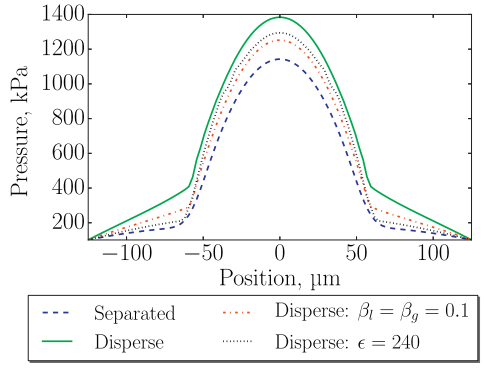
Several factors have been found to influence the disparity between the pressure field maxima, including choice of stabilization parameters, mesh density and liquid permeability parameters. Although increasing domain resolution and decreasing β_l and β_g



(a) Liquid volume vs. time



(b) Pressure after 400 µs



(c) Pressure after 800 µs. Here ϵ is elements per domain edge (as in Table 1)

Fig. 8. Direct comparison of volume conservation characteristics and predicted pressure in separated flow (level-set) and two-mass balance flow models in the single drop case. Subfigure (c) demonstrates the sensitivity of the pressure field to decreasing transition lengths and increasing domain resolution.

all lead to better agreement between pressure predictions, exact agreement between models is not expected in any case.

3.3. Flow visualization

A flow visualization of drop merger was captured with high speed video as pictured in Figs. 4 and 9. A square lattice pattern of 6 pL drops was ink-jetted onto a flat glass plate substrate. A second glass plate was then lowered onto the drops with a displacement-controlled ram. As the gap closes, the pressure becomes very high and plate motion control is switched to constant force while the gas pockets dissolve. The apparatus is surrounded by atmospheric gas at ambient pressure and when fluid reaches the edge of the plate (top boundary in Fig. 9) its motion is arrested by contact line pinning.

Two assumptions about the experiment are made to estimate parameters that are needed for the simulation. The first is that each drop is a spherical cap which sets a relationship between drop volume, contact angle and standing drop radius. The second is that the contact angle is 7.5° which sets the standing drop radius of 39µm. The ratio of lattice spacing to the standing drop radius, measured by pixel ratio, determines the lattice spacing to be 203µm.

The frames of the video are analyzed with computer vision software OpenCV (Bradski, 2000)[4]. A combination of Gaussian blurring and Otsu's binarization[5] is used for each frame to produce an image showing liquid in black and gas in white. Saturation is calculated as the ratio of black pixels to the total number of pixels in a given lattice element. Because the interface is smeared as a result of the video processing, there is uncertainty in the number of black pixels, on the order of the number of pixels along the interface length. This saturation dependent uncertainty is maximized when $S = \pi/4$ at roughly 2.5% of the number of pixels in the lattice element. The gap thickness is subsequently computed from this measured saturation by Eq. (18) because the liquid is incompressible. The saturation predictions and calculated gap thicknesses are only valid after the gap thickness is 1350nm because the drops do not fully bridge the gap until this point.

A simulation of 16 water drops surrounded by CO_2 using a modified height function is compared to the flow visualization experiment. The simulated gap thickness function is stretched in time in order to achieve simulation stability.

Fig. 9 shows the gap thickness as determined from the experiment, a least-squares fit of a hyperbolic tangent function to the calculated gap thickness as well as the smoothed, simulated height function. The oscillation in the experimental data curve right before gap thickness reaches 1350 nm stems from transformation of the drops from spherical sections rising from the substrate to cylinders that bridge the gap. Since the time scales differ between simulation and experiment, the visual comparisons are made at specific gap thicknesses. Inset into Fig. 9 are direct visual comparisons between simulation and experiment at four different gap thicknesses, demonstrating qualitative prediction of the two-phase flow.

This section shows that key process features observed in experimental images are captured with a multidrop simulation. As the gap closes from 1350 nm, the drops expand from the their centers until they merge at 186 nm gap thickness. The pockets of trapped gas compress and dissolve until the gap is occupied by liquid only at a gap of 146 nm. This experimental comparison, while not true validation of the model, demonstrate that the disperse, two-fluid approach is able to capture the gross features of the flow.

3.4. Exemplar device

In NIL processes, templates may contain a variety of features positioned to create complex patterns possessing several tiers of feature density (e.g. transistors are organized to form logic gates that are in turn grouped to create processors). The approach developed here can be adapted to model varied patterns of drops for such purpose. For example, Fig. 10a illustrates a mask for use as a conductive layer etch barrier to be employed in the fabrication of a transistor array. In the center of the pattern is the array of the conductive paths for planar transistors (Choi et al., 2015) that might be used to drive an array of LEDs in a display. The contact

[4] http://opencv.org/.

[5] https://docs.opencv.org/3.3.1/d7/d4d/tutorial_py_thresholding.html.

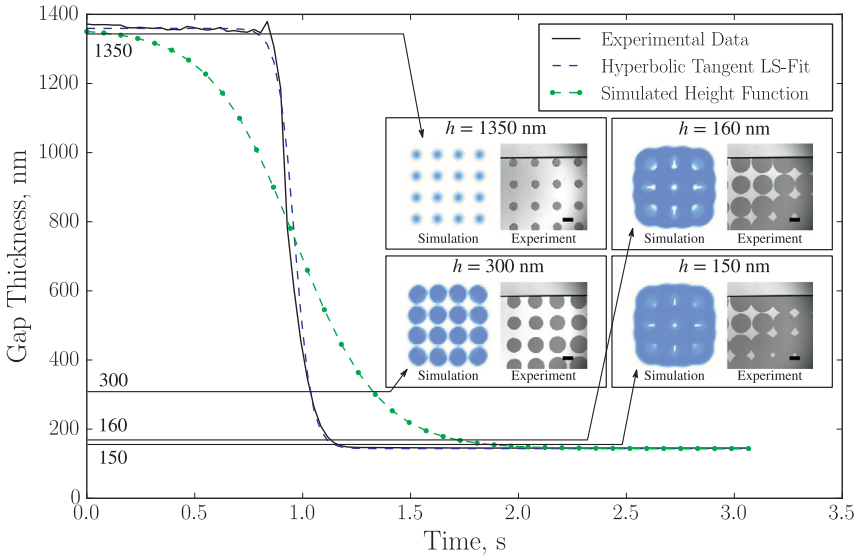
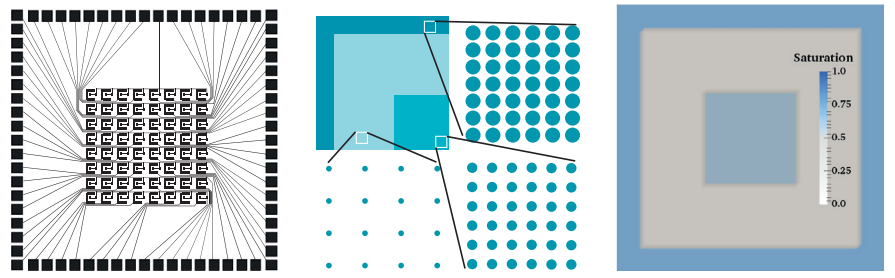


Fig. 9. Plot of gap thickness vs. time for experiment, curve-fit and simulation. Inset are direct visual comparisons at heights marked by parallel lines from the gap thickness axis. Scale bars represent 100 μm.



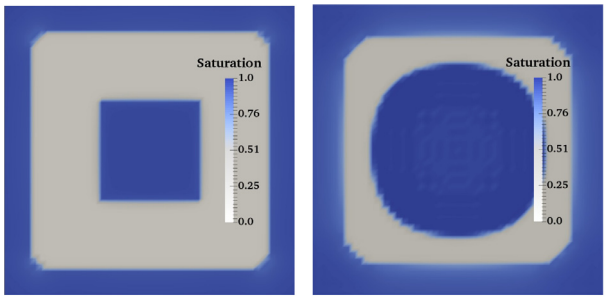
(a) Conductive layer for planar transistor array (b) Varying drop patterns required by varying imprint patterns (c) Saturation field initialization

Fig. 10. Single-layer pattern, commensurate drop patterns and simulation initialization for transistor array.

pads at the edge and the block of transistor elements in the center have a much greater feature density and therefore require more liquid, i.e. a more dense drop pattern than the wire paths connecting the two regions, depicted in Fig. 10b. To demonstrate how J-FIL processes with varying drop densities can be simulated, the saturation field is initialized with corresponding saturation levels, as shown in Fig. 10c. Atmospheric pressure and free outflow conditions are applied at the boundary.

The regions of higher saturation fill first, and as the model does not yet account for patterned templates, once the gap is filled, the liquid must flow into adjacent unsaturated regions. Fig. 11a shows that all the gas in the central and edge regions has escaped or dissolved while the low-liquid-density wiring region has not yet completely filled. Fig. 11b show that as the simulation proceeds, the fully saturated central and edge regions begin to flood the wiring region.

This example demonstrates the capture of two-phase lubrication flow for the class of applications for which the model is intended. In future work, predictive capabilities will be expanded by the addition of physics of substrate/template deformation, capillary action, and fluid flow into the template. Structural mechanics can be incorporated by coupling to membrane mechanics (Carvalho, 2003) or plate mechanics To predict flow anisotropy observed in experiments like those of Zhang et al. (2016) this gap flow model may be coupled with a template flow model based on porous media flow theory as described by Roberts and Schunk (2014). Pattern flow effects of increased dissolution and capillary action could be captured by extension of the porous flow model by the approach to dissolution described by Chauhan et al. (2009).



(a) Filled transistor region (b) Liquid flowing from saturated to unsaturated regions

Fig. 11. Saturation rises to fill the closing gap before flowing outward from a full gap.

4. Conclusion

A model for two-phase flow in a closing gap that is suitable for simulating drop merger of drop-dispensed nanoimprint lithography processes has been presented. The model is implemented in a continuum finite-element code using two-dimensional

shell elements to capture flow. This implementation enables the application of lubrication theory to problems in which incompressible liquids interact with compressible, dissolvable gases. The model demonstrates qualitative prediction of closing gap drop merger by capturing evolution of interfaces in addition to gas trapping/dissolution behaviors that are congruent with video captured of the process.

Conservation of mass and momentum are applied to each phase individually and interaction between phases occurs through the saturation-based relative permeabilities. This disperse, two-fluid flow model enables the simulation of large area, multiphase lubrication processes by reducing the computational complexity, and hence, computation time, with reduced-order models that allow reduction of required spatial resolution. Variation of the dispensed drop pattern, (i.e. spatial variation of drop size, lattice spacing and lattice type) can be addressed in the dissolution model, however these variations are not reflected in the relative permeability model, representing one of the many research pursuits for future work.

Avenues for future work include exploration of alternative gas compression models that may be more appropriate for the length scales and pressures involved in these NIL flows. As the dissolution model is formulated specifically for disperse, two-fluid flows, re-tailoring of the dissolution model may improve the accuracy of discrete drop simulations, although the current model adequately portrays gas trapping and subsequent gas dissolution when diffuse interfaces are included. Improvement of the clipping method will reduce total computation time for both disperse and separated flow simulations. A more detailed study of energy landscape with respect to liquid viscosity, gas properties, RLT and gap closure rate will advance the understanding of the role of gas in drop merger during NIL. Incorporation with thin-shell structural mechanics will extend simulation capabilities to enable prediction of residual layer thickness variations driven by capillary action and coupled deformation of substrate and template. To conclude, this work serves as a building block to be used in the development of more advanced models that will enable predictive simulations to guide the design of drop-dispensing nanoimprint processes.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijmultiphaseflow.2018.03.014.

References

Almqvist, A., Dasht, J., 2005. The homogenization process of the reynolds equation describing compressible liquid flow. Tribol. Int. 39, 994–1002.
Bacri, J.-C., Rakotomalala, N., Salin, D., Woumèni, R., 1992. Miscible viscous fingering experiments versus continuum approach. Phys. Fluids A 4, 1611–1619.
Bear, J., 1972. Dynamics of fluids in porous media. Dover.
Bradford, S.F., Katapodes, N.D., 2000. The anti-dissipative, non-monotone behavior of petrov-galerkin upwinding. Int. J. Numer. Methods Fluids 33, 583–608.
Bradski, G., 2000. The opencv library. Dr. Dobb's Journal of Software Tools. accessed 2017-07-27.
Brennen, C.E., 2005. Fundamentals of Multiphase Flows. Cambridge University Press. Accessed 2017-07-27.
Carvalho, M.S., 2003. Elastohydrodynamics of tensioned web roll coating process. Int. J. Multiphase Flow 41, 561–576.
Chauhan, S., Palmieri, F., Bonnecaze, R.T., Wilson, C.G., 2009. Feature filling modeling for step and flash inprint lithography. J. Vacuum Sci. Technol. B 27, 1926–1932.
Choi, Y., Park, W.-Y., Kang, M.S., Lee, J.-Y., Kim, Y.-H., Cho, J.H., 2015. Monolithic metal oxide transistors. ACS NANO 9, 4288–4295.
Colburn, M., Choi, B.J., Sreenivasan, S., Bonnecaze, R.T., Wilson, C.G., 2004. Ramifications of lubrication theory on imprint lithography. Microelectron. Eng. 75, 321–329.
Cueto-Felgueroso, L., James, R., 2014. A phase-field model of two-phase hele-shaw flow. J. Fluid Mech. 758, 522–552.
Deen, W.M., 1998. Analysis of Transport Phenomena. Oxford University Press.
Fourar, M., Bories, S., 1995. Experimental study of air-water two-phase flow through a fracture (narrow channel). Int. J. Multiphase Flow 21, 621–637.
Fourar, M., Bories, S., Lenormand, R., 1992. Experimental study of two-phase flow in rough fractures. In: Seventeenth Workshop on Geothermal Reservoir Engineering, pp. 215–218.
Fourar, M., Moyne, C., Lenormand, R., 1998. A theoretical model for relative permeabilities in two-phase flow in a fracture. In: Burganos, V.N., Karatzas, G.P., Payatakes, A.C., Brebbia, C.A., Pinder, G.F. (Eds.), Computational Methods in Water Resources XII, pp. 159–166.
Haisma, J., Verheijen, M., van den Heuvel, K., van den Berg, J., 1996. Mold-assisted nanolithography: a process for reliable pattern replication. J. Vac. Sci. Technol., B 14, 4124–4128.
Harrison, W.J., 1913. The hydrodynamical theory of lubrication with special reference to air as a lubricant. Trans. Cambridge Philosoph. Soc. 22, 39–54.
Hele-Shaw, H.S., 1898. Stream-line motion of a viscous film. Report of the British Association for the Advancement of Science 136–142.
Hiroshima, H., Hirai, M.N.Y., Matsui, S., 2013. Innovative nanoimprint lithography using pfp condensible gas. J. Photopolym. Sci. Technol. 26, 87–96.
Jain, A., Spann, A., Bonnecaze, R.T., 2016. Effect of droplet size, droplet placement, and gas dissolution on throughput and defect rate in uv nanoimprint lithography. J. Vac. Sci. Technol., B 35, 011602-1–011602-12.
Jain, A., Spann, A., Cochrane, A., Schunk, P.R., Bonnecaze, R.T., 2017. Fluid flow in uv nanoimprint lithography with patterned templates. Microelectron. Eng. 173, 62–70.
Leonard, B.P., Niknafs, H.S., 1991. Sharp monotonic resolution of discontinuities without clipping of narrow extrema. Comput. Fluids 19, 141–154.
Li, W.-D., Wu, W., Williams, R.S., 2012. Combined helium ion beam and nanoimprint lithography attains 4nm half-pitch dense patterns. J. Vac. Sci. Technol., B 30, 06F304–1–06F304–4.
Liang, X., Tan, H., Fu, Z., Chou, S.Y., 2007. Air bubble formation and dissolution in dispensing nanoimprint lithography. Nanotechnology 18, 1926–1932.
Lockhart, R.W., Martinelli, R.C., 1949. Proposed correlation of data for isothermal two-phase two-component flow in pipes. Chem. Eng. Prog. 45 (1), 39–48.
Morse, J., Carter, K., Clark, P., Daniels, R., Du, X., Ernst, J., Gallatin, G.M., Gamota, D., Goyal, A., Guo, J., Hardt, D., Hunter, M., Kadtke, J., Kehagias, N., Low, H.-Y., Niblock, T., Petrzelka, J., Poliks, M., Praino, R., Ranuik, G., Slafer, D., Soles, C., Sreenivasan, S., Sundaram, G., Watkins, J., Whitman, L., Wright, M., 2011. Nanofabrication technologies for roll-to-roll processing. Technical Report. NIST–National Nanomanufacturing Network.
Papanastasiou, T.C., Malamataris, N., Ellwood, K., 1992. A new outflow boundary condition. Int. J. Numer. Methods Fluids 14, 587–608.
Peng, L., Wu, H., Yi, P., Lai, X., 2016. Study on bubble defect in roll-to-roll uv imprinting process for micropyramid arrays. ii. numerical study. J. Vacuum Sci. Technol. B 34, 2166–2746.
Reddy, S., Bonnecaze, R.T., 2005. Simulation of fluid flow in the step and flash imprint lithography process. Microelectron. Eng 82, 60–70.
Reynolds, O., 1886. On the theory of lubrication and its application to mr. beauchamp tower's experiments, including an experimental determination of the viscosity of olive oil. Philosoph. Trans. Royal Soc. London 177, 157–234.
Roberts, S.A., Noble, D.R., Benner, E.M., Schunk, P.R., 2013. Multiphase hydrodynamic lubrication flow using a three-dimensional shell finite element model. Comput. Fluids 87, 12–25.
Roberts, S.A., Schunk, P.R., 2014. A reduced-order model for porous flow through thin, structured materials. Int. J. Multiphase Flow 67, 25–36.
Romm, E.S., 1966. Fluid flow in fractured rocks. Nedra Publishing House. [english trans. William R. Blake (1972)].
Schift, H., 2008. Nanoimprint lithography: an old story in modern times? a review. J. Vac. Sci. Technol., B 26, 458–480.
Schmid, G.M., Brooks, C., Ye, Z., Johnson, S., LaBrake, D., Sreenivasan, S.V., Resnick, D.J., 2009. Jet and flash imprint lithography for the fabrication of patterned media drives. In: Proceedings of SPIE. Photomask Technology.
Schunk, P.R., Sackinger, P.A., Rao, R.R., Chen, K.S., Baer, T.A., Labreche, D.A., Sun, A.C., Hopkins, M.M., Subia, S.R., Moffat, H., Secor, R.B., Roach, R.A., Wilkes, E.D., Noble, D.R., Hopkins, P.L., Notz, P.K., Roberts, S.A., 2013. Goma 6.0 - A Full-Newton Finite Element Program for Free and Moving Boundary Problems with Coupled Fluid/Solid Momentum, Energy, Mass and Chemical Species Transport: User's Guide. Technical Report. Sandia National Laboratories.

Chou, S.Y., Krauss, P.R., Renstrom, P.J., 1995. Imprint of sub-25 nm vias and trenches in polymers. Appl. Phys. Lett. 67, 3114.
Stokes, G.G., 1898. Mathematical proof of the identity of the stream lines obtained by means of a viscous film with those of a perfect fluid moving in two dimensions. Report of the British Association for the Advancement of Science 143–144.
Taylor, H., 2016. Defectivity prediction for droplet-dispensed uv nanoimprint lithography, enabled by fast simulation of resin flow at feature, droplet, and template scales. In: Proceedings of SPIE, 9777. Alternative Lithographic Technologies VIII.
Usuki, K., Wakamatsu, S., Oomatsu, T., Hattori, A., Tarutani, S., Kodama, K., Tanabe, H., Shitabatake, K., 2012. Design considerations for ultraviolet–nanoimprint lithography resists. J. Micro/Nanolithogr. MEMS MOEMS 11, 033008-1–033008-8.
Wang, Q., Hiroshima, H., 2010. Effects of environmental gas in uv nanoimprint on the characteristice of uv-curable resin. Jpn. J. Appl. Phys. 49 06GL04-1–06GL04-4.
Wyckoff, R.D., Botset, H.G., 1936. The flow of gas-liquid mixtures through unconsolidated sands. J. Appl. Phys. 7, 325–345.
Zhang, W., Fletcher, B., Thompson, E., Liu, W., Stachowiak, T., 2016. High throughput jet and flash imprint lithopgraphy for semiconductor memory applications. In: Proceedings of SPIE. Alternative Lithographic Technologies VIII.
Zienkiewicz, O.C., Taylor, R.L., 2000. The Finite Element Method. Butterworth-Hienemann.